Lecture plan · for the lecturer

Heuristics and simulated annealing

Optimization · University · Lecture 3

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Use local search.
- Explain annealing.
- Evaluate heuristics.

Activity: Every row a seed

1. Warm-up (5 min): دقيق أم سريع؟
2. Search (15 min): Run local search and see where it gets stuck.
3. Anneal (20 min): Each row runs annealing with its own seed and reports the final cost.
4. Evaluate (15 min): Plot all results and compare with the known optimum.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How wide was the spread?
- What did cooling faster do?
- How will you require evaluation?

Exit ticket

- Write the evaluation rule you will require for heuristics.

Try it in your next lecture
Give each row of students a different random seed and the same timetable problem; each row runs annealing and reports its final cost, then plot all results and discuss the spread.

Seminar sheet · for students

Heuristics and simulated annealing

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Name _____ Class _____ Date _____

1 **At T = 2, a move worse by 2 is accepted with probability about:**

◯ 37% ◯ 14% ◯ 100%

2 **Compute acceptance.**

A move worse by Δ = 3 at T = 3 is accepted with probability e^(_____), about _____ %.

Word bank: -1 · -3 · 5 · 37

3 **Which make a heuristic comparison fair? Choose all that apply.**

☐ Several seeds per method ☐ Equal time budgets ☐ Tuning on the comparison instances

☐ Reporting gaps to a lower bound

4 **Simulated annealing with a geometric cooling schedule is guaranteed to find the global optimum.**

◯ True ◯ False

5 **Match each temperature to the search behaviour.**

Very high T	• • • •	Behaves like pure local search
Moderate T	• • • •	Explores but prefers better moves
T near zero	• • • •	Close to a random walk

✎ **How do your students currently report heuristic results, and what would convince you a result is good?**